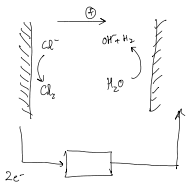
Lecture 2.

Monday, 16 January 2023 8:12 AM

Electroneutrality!




Electrolyte: phase thru which ions can move
1) Can be liquid (aq or organic)
2) Polymeric (Nafion)
3) Solid (ceramic w/ oxide vacancies or solid-oxide electrolyte)

Ex: "Chlor-Alkali Process" OIL RIG

$2\,Cl^-_{(aq)} \rightarrow Cl_2(g) + 2e^-$ (Oxidation)

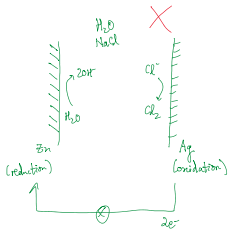
$2H_2O + 2e^- \rightarrow 2OH^- + H_2(g)$ (Reduction)

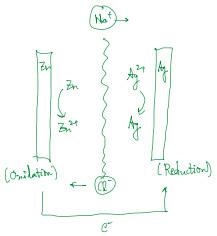
$2NaCl_{(aq)} + 2H_2O \rightarrow Cl_2(g) + 2NaOH + H_2(g)$

Shorthand: $Pt \mid Cl^-_{(aq)} \parallel NaOH_{(aq)} \mid H_2, Cl_2(g) \mid Pt_{(s)}$

↑ Phase boundary ↑ transition to different electrolyte composition

Ex: Alessandro Volta: first battery in 1790's




Wind + Solar → enable electrochemical processes in manufacturing to take place

$Ag_2O + H_2O + 2e^- \rightarrow Ag + 2OH^-$

$Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$

Note: Valency $2+ \rightarrow 3+$ } Oxidation (Lost an $e^-$)

∴ Number increasing → oxidation

$e^-$ being released (right hand side) → Oxidation